Table 56. ADC accuracy(1)(2)

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
ET	Total unadjusted error	$V_{DDA} = V_{REF+} = 3$ V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, $T_A = 25°C$	-	±3	±4	LSB
		2 V < $V_{DDA} = V_{REF+}$ < 3.6 V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, T_A = entire range	-	±3	±6.5	
EO	Offset error	$V_{DDA} = V_{REF+} = 3$ V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, $T_A = 25°C$	-	±1.5	±2	
		2 V < $V_{DDA} = V_{REF+}$ < 3.6 V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, T_A = entire range	-	±1.5	±4.5	
EG	Gain error	$V_{DDA} = V_{REF+} = 3$ V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, $T_A = 25$ °C	-	±3	±3.5	
		2 V < $V_{DDA} = V_{REF+}$ < 3.6 V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, T_A = entire range	-	±3	±5	
ED	Differential linearity error	$V_{DDA} = V_{REF+} = 3$ V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, $T_A = 25$ °C	-	±1.2	±1.5	
		2 V < $V_{DDA} = V_{REF+}$ < 3.6 V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, T_A = entire range	-	±1.2	±1.5	
EL	Integral linearity error	$V_{DDA} = V_{REF+} = 3$ V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, $T_A = 25$ °C	-	±2.5	±3	
		2 V < $V_{DDA} = V_{REF+}$ < 3.6 V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, T_A = entire range	-	±2.5	±3	
ENOB	Effective number of bits	$V_{DDA} = V_{REF+} = 3$ V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, $T_A = 25$ °C	10.1	10.2	-	bit
		2 V < $V_{DDA} = V_{REF+}$ < 3.6 V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, T_A = entire range	9.6	10.2	-	
SINAD	Signal-to-noise and distortion ratio	$V_{DDA} = V_{REF+} = 3$ V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, $T_A = 25$ °C	62.5	63	-	dB
		2 V < $V_{DDA} = V_{REF+}$ < 3.6 V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, T_A = entire range	59.5	63	-	
SNR	Signal-to-noise ratio	$V_{DDA} = V_{REF+} = 3$ V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, $T_A = 25$ °C	63	64	-	dB
		2 V < $V_{DDA} = V_{REF+}$ < 3.6 V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, T_A = entire range	60	64	-	
THD	Total harmonic distortion	$V_{DDA} = V_{REF+} = 3$ V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, $T_A = 25$ °C	-	-74	-73	dB
		2 V < $V_{DDA} = V_{REF+}$ < 3.6 V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, T_A = entire range	-	-74	-70	

1. Evaluated by characterization – Not tested in production.
2. ADC DC accuracy values are measured after internal calibration.